

K-디지털 트레이닝

FrontController

[KB] IT's Your Life

- ✓ ex06 프로젝트명을 Dynamic Web 프로젝트를 생성하세요.

✓ 다음 설정으로 FrontControllerServlet을 작성하세요.

- 패키지: org.scoula.ex06
- url 맵핑: /
- 메서드
 - init()
 - doGet()
 - doPost()

FrontControllerServlet.java

```
@WebServlet(name = "frontControllerServlet", value = "/")
public class FrontControllerServlet extends HttpServlet {

    public void init() {
    }

    public void doGet(HttpServletRequest request, HttpServletResponse response) throws IOException, ServletException {
    }

    public void doPost(HttpServletRequest request, HttpServletResponse response) throws IOException, ServletException {
        doGet(request, response);
    }

}
```

- ✓ 다음 조건으로 Command 인터페이스를 정의하세요.
 - 패키지: `org.scoula.ex06.command`
 - 메서드
 - `String execute(HttpServletRequest request, HttpServletResponse response)` throws `IOException`

Command.java

```
package org.scoula.ex06.command;

import javax.servlet.ServletException;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import java.io.IOException;

public interface Command {
    String execute(HttpServletRequest request, HttpServletResponse response) throws IOException;
}
```

- ✓ FrontControllerServlet의 필드로 GET요청 처리 맵과 POST 요청 처리 맵을 추가하세요.
 - init() 메서드에서 인스턴스를 생성함

FrontControllerServlet.java

```
@WebServlet(name = "frontControllerServlet", value = "/")
public class FrontControllerServlet extends HttpServlet {
    Map<String, Command> getMap;
    Map<String, Command> postMap;

    public void init() {
        getMap = new HashMap<>();
        postMap = new HashMap<>();
    }
    ...
}
```


- ✓ 요청 URL에서 요청명을 찾아 리턴하는 메서드를 작성하세요.
 - `private String getCommandName(HttpServletRequest request)`
- ✓ 요청 URL에서 요청명을 추출한 후, 요청 메서드에 따라 해당 Command를 찾아 리턴하는 메서드를 작성하세요.
 - `private Command getCommand(HttpServletRequest request)`

FrontControllerServlet.java

```
@WebServlet(name = "frontControllerServlet", value = "/")
public class FrontControllerServlet extends HttpServlet {
    ...
    private String getCommandName(HttpServletRequest request) {
        String requestURI = request.getRequestURI();
        String contextPath = request.getContextPath();
        return requestURI.substring(contextPath.length());
    }

    private Command getCommand(HttpServletRequest request) {
        String commandName = getCommandName(request);

        Command command;
        if(request.getMethod().equalsIgnoreCase("GET")) {
            command = getMap.get(commandName);
        } else {
            command = postMap.get(commandName);
        }

        return command;
    }
}
```

- ✔ 요청을 분석하여 추출된 Command를 실행하는 메서드를 작성하세요.
 - `private void execute(Command command, HttpServletRequest request, HttpServletResponse response)`
 - Command 실행으로 리턴된 뷰이름은 `viewName` 변수로 저장해 둔다.

FrontControllerServlet.java

```
@WebServlet(name = "frontControllerServlet", value = "/")
public class FrontControllerServlet extends HttpServlet {
    ...

    private void execute(Command command, HttpServletRequest request, HttpServletResponse response)
        throws IOException, ServletException {

        String viewName = command.execute(request, response);
    }
}
```

✓ doGet 메서드를 다음과 같이 작성하세요.

```
@WebServlet(name = "frontControllerServlet", value = "/")
public class FrontControllerServlet extends HttpServlet {
    ...

    public void doGet(HttpServletRequest request, HttpServletResponse response) throws IOException, ServletException {
        Command command = getCommand(request);
        if(command != null) {
            execute(command, request, response);
        } else { // 404 에러 처리
        }
    }
}
```